Maria Eleni Zoumpoulidi

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I am a Master's student and Researcher interested in LLMs' reasoning, interpretability and explainability.

EDUCATION

09/2026 - 09/2027 **Imperial College London**, London, UK
- Master of Research (MRes) in AI and Machine Learning

09/2019 - 03/2025 **National Technical University of Athens (NTUA),** Athens, Greece
- Diploma (MEng) in Electrical and Computer Engineering (5-year Integrated Master's degree, 300 ECTS)
  - Grade: 7.98/10.00 "Very Good"
  - Relevant Coursework: Pattern Recognition (10/10), Speech & Natural Language Processing (9/10), AI (8/10), Neural Networks and Deep Learning (7/10), Computability & Complexity (10/10)
- Diploma thesis: **"BloomWise: Enhancing Problem-Solving capabilities of LLMs using Bloom's-Taxonomy-inspired prompts",** link (supervisor: prof. Alexandros Potamianos (Scholar)) (10/10)

RESEARCH AND PROFESSIONAL EXPERIENCE

03/2025 – 06/2026 **Associate Researcher,** Institute for Language and Speech Processing, Athena Research Center, Greece
- Conducted research on LLMs' reasoning, interpretability, and explainability.
- **Led and first-authored** *BloomWise* (improving LLMs' reasoning through cognitively-inspired in-context learning, MathNLP@EMNLP2025), *BloomXplain* (a framework and benchmark for pedagogically sound LLM-generated explanations, LLM-eval@NeurIPS2025), and a work on cognitive profiling of LLMs (under review);
- **Co-led and co-first-authored** two works on interpretability and faithfulness regarding mathematical solvability detection (one accepted at MRL@EMNLP26, one under review).
- Performed LLMs' fine-tuning and inference on HPC systems.
- Developed retrieval-augmented generation (RAG) pipelines.

07/2022 - 08/2022 **Intelligent Decisions Intern**, Qualco (international fintech solutions provider), Greece
- Responsibilities: production of prototype explainers for NLP-based ML predictive models

PUBLICATIONS AND PREPRINTS

- **BloomWise: Enhancing Problem-Solving capabilities of Large Language Models using Bloom's Taxonomy-Inspired Prompts** (paper)
  Maria-Eleni Zoumpoulidi, Georgios Paraskevopoulos, Alexandros Potamianos
  MathNLP Workshop @ EMNLP 2025 (archival, oral presentation)
- **BloomXplain: A Framework and Benchmark Dataset for Pedagogically Sound LLM-Generated Explanations Based on Bloom's Taxonomy** (paper)
  Maria-Eleni Zoumpoulidi, Eleni Batsi, Georgios Paraskevopoulos, Vassilis Katsouros, Alexandros Potamianos
  LLM-eval Workshop @ NeurIPS 2025 (non-archival, poster)
- **More Capable, Less Faithful: A Multilingual Analysis of Mathematical (Un)Solvability Detection in LLMs** (paper)
  Maria-Eleni Zoumpoulidi*, Nikolaos Xiros*, Georgios Paraskevopoulos (co-first authors, randomized order)
  MRL Workshop @ EMNLP 2026
- **Knowledge Knows, Verbalization Tells: Disentangling Latent Directions for Mathematical Solvability in LLMs** (paper)
  Nikolaos Xiros*, Maria-Eleni Zoumpoulidi*, Georgios Paraskevopoulos, preprint (co-first authors, randomized author order)
  Arxiv preprint
- **Cognitive Profiling of LRMs' Reasoning Traces Using Bloom's Taxonomy** (paper)
  Maria-Eleni Zoumpoulidi, Georgios Paraskevopoulos, Alexandros Potamianos
  Arxiv preprint
- **Beyond Magnitude: Contrastive Routing for Modular Mixture-of-Experts** (paper)
  Nikolaos Xiros, Dimitrios Damianos, Maria-Eleni Zoumpoulidi, Leon Voukoutis Vassilis Katsouros Georgios Paraskevopoulos
  EMNLP 2026

VOLUNTEERING EXPERIENCE

- Coordinator of the Machine Learning team, IEEE NTUA Student Branch, Computer Society Chapter, Apr 2023 - Oct 2023

EXTRACURRICULAR ACTIVITIES

- Erasmus+ Instructor: AI & Digital Transformation in Higher Education, Jun 2025
- GoCERN Summer School (CERN-NTUA-Arsakeio Tositseio School) , Jun 2019
- Participation in the Athens Model of United Nations (MUN), Mar 2017

SKILLS

- Software: Python, C++, SQL, Bash, Git, Operating Systems
- Machine Learning: PyTorch, HuggingFace, NumPy, Pandas

LANGUAGES

- **English** (fluent-Proficiency), **French** (independent user-Delf B2), **Greek** (native)